Lecture 9

In this lecture notes we prove the *invertibility theorem* which shows in particular that any invertible matrix is row equivalent to the identity matrix (of the same size) and that it can be written as a product of elementary matrices.

Matrix invertibility theorem

Lemma 1. *Let U be an $n \times n$ matrix in row echelon form. If $u_{ii} \neq 0$ for $1 \leq i \leq n$, then U is row equivalent to I_n.*

Sketch. Since U is in row echelon form, it will have form:

$$U = \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ 0 & 0 & \cdots & u_{3n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_{nn} \end{pmatrix}$$

Proceeding similar to Gaussian elimination, we could use row operations involving the last row to make all entries above u_{nn} zero, and finally multiply the last row by u_{nn}^{-1}. We can then more to the second last column, third last column etc performing similar operations until we get the identity matrix.

Lemma 2. *Let A and B be $m \times n$ row equivalent matrices, then $A\boldsymbol{x} = \boldsymbol{0}$ and $B\boldsymbol{x} = \boldsymbol{0}$ have exactly the same solutions.*

Proof. Suppose A is row equivalent to B, then there exists operations $e_1, e_2, \ldots, e_k$ such that $A = e_k \ldots e_2 e_1(B)$, and so we have elementary matrices $E_1, E_2, \ldots, E_k$ such that $A = E_k \ldots E_2 E_1 B$ and $B = E_1^{-1} E_2^{-1} \ldots E_k^{-1} A$. Therefore, if $\mathbf{u}$ is a solution for $B\mathbf{x} = \mathbf{0}$, that is, $B\mathbf{u} = \mathbf{0}$, we have $A\mathbf{u} = E_k \ldots E_2 E_1 B\mathbf{u} = E_k \ldots E_2 E_1 \mathbf{0} = \mathbf{0}$, and so $\mathbf{u}$ is also a solution for $A\mathbf{x} = \mathbf{0}$, similary if $\mathbf{u}$ is a solution for $A\mathbf{x} = \mathbf{0}$, we have $B\mathbf{u} = E_1^{-1} E_2^{-1} \ldots E_k^{-1} A\mathbf{u} = \mathbf{0}$ □

Definition 24. *A matrix A is said to be singular if the equation $A\boldsymbol{x} = \boldsymbol{0}$ has more than one solution. Otherwise A is said to be non-singular, this is the case when equation $A\boldsymbol{x} = \boldsymbol{0}$ has only one solution, namely, the trivial solution.*

Examples Note that the first row on matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}$ is the sum of the second and third row, thus row reducing it to row echelon form one will get that it will have at most two pivots, and having three variables we get that the equation $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions. That is, the above matrix A is singular. (check the details).

Matrix $B = \begin{pmatrix} 1 & 4 \\ 1 & 2 \end{pmatrix}$ can be row reduced to triangular form and thus equation $B\mathbf{x} = \mathbf{0}$ has a unique solution and so B is non-singular.

Theorem 1. *(Invertibility theorem) Let A be an $n \times n$ matrix. Then the following statements are equivalent:*

(a) A is invertible;
(b) $A\boldsymbol{x} = \boldsymbol{b}$ has a unique solution for every vector $\boldsymbol{b}$;
(c) A is non-singular;
(d) A is row equivalent to I_n;
(e) A is a product of elementary matrices.

Proof. We shall prove the above theorem by proving implications $(a) \implies (b) \implies (c) \implies (d) \implies (e) \implies (a)$.
$(a) \implies (b)$: Suppose A is invertible, then A^{-1} exists and we have that $\mathbf{x} = A^{-1}\mathbf{b}$ is a solution, indeed, $A(A^{-1}\mathbf{b}) = (AA^{-1})\mathbf{b} = I_n\mathbf{b} = \mathbf{b}$, now suppose $\mathbf{u}$ and $\mathbf{v}$ are solutions, we have that $A\mathbf{u} = \mathbf{b} = A\mathbf{v}$ from which we get $\mathbf{u} = \mathbf{v}$ by left multiplying by A^{-1} both sides and so our solution is unique.
$(b) \implies (c)$: Suppose $A\mathbf{x} = \mathbf{b}$ has a unique solution for every vector $\mathbf{b}$, in particular taking $\mathbf{b} = \mathbf{0}$, we have that $A\mathbf{x} = \mathbf{0}$ has a unique solution and thus A is non-singular.
$(c) \implies (d)$: Suppose $A\mathbf{x} = \mathbf{0}$ has a unique solution, then we should show that we can row reduce A to I_n. First we can reduce A to a matrix U in row echelon form.

$$U = \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ 0 & 0 & \cdots & u_{3n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_{nn} \end{pmatrix}$$

If we show that for $1 \leq i \leq n$, u_{ii} is non-zero, then by the first lemma we have that we can row reduce U to I_n. We now prove by contradition the following statement:

$$\text{If } A\mathbf{x} = \mathbf{0} \text{ has a unique solution, then } u_{ii} \neq 0 \text{ for } 1 \leq i \leq n.$$

Now, suppose on the contrary that one of the u_{ii}'s is zero, then since U is a square matrix in row echelon form u_{nn} must also be zero. But this means that we have at most $n-1$ pivots and so at least one variable must be free. Thus equation $U\mathbf{x} = \mathbf{0}$ has infinitely many solutions and therefore so is $A\mathbf{x} = \mathbf{0}$ because A and U are row equivalent. This means that A is singular (a contradiction). Therefore we must have that $u_{ii} \neq 0$ for $1 \leq i \leq n$.
$(d) \implies (e)$: Suppose that A is row equivalent to I_n, then there exists elementary matrices $E_1, E_2, \ldots, E_k$ such that $A = E_k \ldots E_2E_1I_n$, that is $A = E_k \ldots E_2E_1$ and so A is a product of elementary matrices.
$(e) \implies (a)$: Suppose A is a product of elementary matrices, since elementary matrices are invertible, the we have that A is a product of invertible matrices and thus A is invertible.

□